5. Spaces of integrable functions

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Definition 5.1: Metric Space Let V be a set. A function $d : V \times V \to \mathbb{R}$ is a *metric* on V (making (V, d) a metric space) if for all $x, y, z \in V$:

(i) $d(x, y) \geq 0$,

(ii) $d(x, y) = 0 \iff x = y$,

(iii) $d(x, y) = d(y, x)$,

(iv) $d(x, z) \leq d(x, y) + d(y, z)$.

Definition 5.2: Normed Vector Space Let V be a vector space over $\mathbb{R}$ (or $\mathbb{C}$). A function $\|\cdot\| : V \to \mathbb{R}$ is a *norm* on V if for all $x, y \in V$ and scalars α:

(i) $\|x\| \geq 0$,

(ii) $\|x\| = 0 \iff x = 0$,

(iii) $\|\alpha x\| = |\alpha| \, \|x\|$,

(iv) $\|x + y\| \leq \|x\| + \|y\|$.

Note: Every norm induces a metric by setting $d(x, y) = \|x - y\|$.

Definition: Equivalence relation on $\mathcal{L}^1(E)$

For $E \in \mathcal{M}$, functions $f, g \in \mathcal{L}^1(E)$ are equivalent $(f \equiv g)$ if $f = g$ a.e. on E.

Lemma 5.2.5 The relation $\equiv$ is an equivalence relation.

Crux. Reflexivity and symmetry are trivial. For transitivity, suppose $f \equiv g$ and $g \equiv h$. The set where they differ satisfies $\{f \neq h\} \subseteq \{f \neq g\} \cup \{g \neq h\}$. Since the union of two null sets is a null set, $f = h$ a.e., meaning $f \equiv h$. □

Definition: The Space $L^1(E)$ and Well-Defined Operations

The quotient space $L^1(E) \triangleq \mathcal{L}^1(E)/\equiv$ consists of equivalence classes $[f]$. Linear operations on these classes are **well-defined** (independent of the chosen representative).

Crux. If $f_1 \equiv f_2$ and $g_1 \equiv g_2$, then $f_1 + g_1 = f_2 + g_2$ except on the union of two null sets. Thus, $f_1 + g_1 \equiv f_2 + g_2$, meaning $[f_1 + g_1] = [f_2 + g_2]$. Similarly, $cf_1 \equiv cf_2$, so $[cf_1] = [cf_2]$. □

Note: Because operations are well-defined, we drop the brackets and treat $L^1(E)$ as a vector space of functions where $f = g$ means equality a.e. The function $\|f\| \triangleq \int_E |f| \, \mathrm{d}m$ is a valid norm on $L^1(E)$ because $\|f\| = 0 \implies f = 0$ a.e., which is exactly the zero element of $L^1(E)$.

Definition 5.3: Cauchy Sequence and Completeness Let V be a vector space with norm $\|\cdot\|_V$. A sequence $\{f_n\} \subset V$ is *Cauchy* if:

$$\forall \epsilon > 0, \exists N \text{ such that } \forall n, m \geq N, \|f_n - f_m\|_V < \epsilon.$$

If every Cauchy sequence converges to an element within V, then V is *complete* (a complete normed vector space is called a Banach space).

Theorem 5.5. *(Riesz-Fischer Theorem for L^1)*
The space $L^1(E)$ is complete.

WARNING: Do not assume that because $\{f_n\}$ is Cauchy in $L^1(E)$, the sequence of real numbers $\{f_n(x)\}$ is pointwise Cauchy for a fixed x (e.g., the "sliding typewriter" counterexample). We must extract a fast-converging subsequence first.

Crux. Let $\{f_n\}$ be Cauchy in $L^1(E)$. We proceed in four steps:

(1) **Fast Subsequence:** Choose a strictly increasing sequence N_k such that $\|f_n - f_m\| < 1/2^k$ for all $n, m \geq N_k$. Thus, $\left\|f_{N_{k+1}} - f_{N_k}\right\| < 1/2^k$.

(2) **Pointwise Limit:** Let $g = \sum_{k=1}^{\infty} \left|f_{N_{k+1}} - f_{N_k}\right|$. By Beppo-Levi (Theorem 4.30), $\int_E g \, \mathrm{d}m = \sum_{k=1}^{\infty} \left\|f_{N_{k+1}} - f_{N_k}\right\| \leq 1$. Since $g \in L^1(E)$, $g < \infty$ a.e. Therefore, the telescoping series $f_{N_1} + \sum_{k=1}^{\infty}(f_{N_{k+1}} - f_{N_k})$ converges absolutely a.e. to some limit function f.

(3) L^1 **Convergence of Subsequence:** For any k_0, $\left|f - f_{N_{k_0}}\right| \leq \sum_{j=k_0}^{\infty} \left|f_{N_{j+1}} - f_{N_j}\right| \leq g$. Since $f_{N_{k_0}} \to f$ pointwise a.e. and is dominated by $g \in L^1(E)$, the Dominated Convergence Theorem yields $\left\|f - f_{N_{k_0}}\right\| \to 0$. (Also, $|f| \leq |f_{N_1}| + g$, so $f \in L^1(E)$).

(4) **Whole Sequence Convergence:** In any metric space, if a Cauchy sequence has a convergent subsequence, the entire sequence converges to the same limit. Thus, $f_n \to f$ in $L^1(E)$.

□

Definition: The Space $L^2(E)$
Let

$$\|f\|_2 \triangleq \left(\int_E |f|^2 \, \mathrm{d}m \right)^{1/2}.$$

We say $f \in \mathcal{L}^2(E)$ if this integral is finite. Similar to the L^1 case, we define the quotient space $L^2(E) \triangleq \mathcal{L}^2(E)/\equiv$ (where $f \equiv g \iff f = g$ a.e.). We treat every equivalence class $[f]$ as a single function f. For the complex case, we denote it by $L^2(E, \mathbb{C})$. It is easily shown that $L^2(E)$ is a vector space.

Theorem 5.6. *(Cauchy-Schwarz Inequality) If $f, g \in L^2(E, \mathbb{C})$, then $fg \in L^1(E, \mathbb{C})$ and*

$$\left| \int_E f\overline{g} \, \mathrm{d}m \right| \le \|fg\|_1 \le \|f\|_2 \, \|g\|_2 \, .$$

Crux. Substituting $h = f\overline{g}$ into $\left|\int_E h \, \mathrm{d}m\right| \le \int_E |h| \, \mathrm{d}m$ immediately gives the first inequality. For the second, we can assume without loss of generality that f and g are real and non-negative. To avoid infinities, truncate the domain to $E_k = E \cap [-k, k]$ and the functions to $f_n = \min(f, n)$ and $g_n = \min(g, n)$. For any $t \in \mathbb{R}$:

$$\int_{E_k} (f_n + tg_n)^2 \, \mathrm{d}m = \int_{E_k} f_n^2 \, \mathrm{d}m + 2t \int_{E_k} f_n g_n \, \mathrm{d}m + t^2 \int_{E_k} g_n^2 \, \mathrm{d}m \ge 0.$$

This is a non-negative quadratic in t, so its discriminant ($b^2 - 4ac$) must be non-positive:

$$4 \left(\int_{E_k} f_n g_n \, \mathrm{d}m \right)^2 - 4 \int_{E_k} f_n^2 \, \mathrm{d}m \int_{E_k} g_n^2 \, \mathrm{d}m \le 0$$
$$\implies \int_{E_k} f_n g_n \, \mathrm{d}m \le \|f_n\|_{L^2(E_k)} \, \|g_n\|_{L^2(E_k)} \le \|f\|_2 \, \|g\|_2 \, .$$

Since $f_n g_n \nearrow fg$ and $E_k \nearrow E$, applying the Monotone Convergence Theorem yields $\|fg\|_1 \le \|f\|_2 \, \|g\|_2$. □

Corollary (Minkowski's Inequality for L^2): Expanding $\|f + g\|_2^2 = \int_E (f+g)(\overline{f + g}) \, \mathrm{d}m$ and applying Cauchy-Schwarz to the cross-term easily yields $\|f + g\|_2 \le \|f\|_2 + \|g\|_2$, confirming $\|\cdot\|_2$ is a valid norm.

Proposition 5.7. *If the set D has finite measure ($m(D) < \infty$), then $L^2(D) \subseteq L^1(D)$.*

Crux. For any $t \ge 0$, the expansion of $(t-1)^2 \ge 0$ yields $2t \le t^2 + 1$, which implies $t \le t^2 + 1$. Substituting $t = |f|$ and integrating over the domain D:

$$\|f\|_1 = \int_D |f| \, \mathrm{d}m \le \int_D (1 + |f|^2) \, \mathrm{d}m = m(D) + \|f\|_2^2 \, .$$

If $f \in L^2(D)$, then $\|f\|_2^2 < \infty$. Since $m(D) < \infty$, the RHS is finite, meaning $\|f\|_1 < \infty$ and $f \in L^1(D)$. □

Alternative Crux (via Cauchy-Schwarz): Applying the Cauchy-Schwarz inequality to $|f|$ and the constant function $g = 1$ on D:

$$\|f\|_1 = \int_D |f| \cdot 1 \, \mathrm{d}m \le \|f\|_2 \left(\int_D 1^2 \, \mathrm{d}m \right)^{1/2} = \|f\|_2 \sqrt{m(D)}.$$

Since both terms on the RHS are finite, $f \in L^1(D)$.

Definition 5.8: Inner Product & Induced Norm For $f, g \in L^2(E, \mathbb{C})$, the *inner product* is defined as $(f, g) \triangleq \int_E f\overline{g} \, \mathrm{d}m$. This induces the standard L^2-norm:

$$\|f\|_2 = \sqrt{(f, f)} = \left(\int_E |f|^2 \, \mathrm{d}m \right)^{1/2}.$$

Proposition 5.9. *The inner product satisfies the following for all $f, g, h \in L^2(E, \mathbb{C})$ and $c \in \mathbb{C}$:*

(i) ***Linearity:*** $(f + g, h) = (f, h) + (g, h)$ *and* $(cf, h) = c(f, h)$.

(ii) ***Conjugate Symmetry:*** $(f, g) = \overline{(g, f)}$.

(iii) ***Positive Definiteness:*** $(f, f) \ge 0$, *with* $(f, f) = 0 \iff f = 0$.

(iv) ***Conjugate Linearity:*** $(f, cg + h) = \overline{c}(f, g) + (f, h)$.

Crux. Properties (i)-(iii) follow immediately from the linearity of the Lebesgue integral and the identity $\overline{\int_E f \, \mathrm{d}m} = \int_E \overline{f} \, \mathrm{d}m$. Property (iv) is a direct consequence of (i) and (ii): $(f, cg + h) = \overline{(cg + h, f)} = \overline{c(g, f) + (h, f)} = \overline{c}(f, g) + (f, h)$. □

Definition 5.10: Inner Product Space & Hilbert Space A vector space H over $\mathbb{C}$ equipped with a map $(\cdot, \cdot) : H \times H \to \mathbb{C}$ satisfying (i)-(iii) of Proposition 5.9 is an *inner product space*. If H is also complete with respect to the induced norm $\|h\| = \sqrt{(h, h)}$, it is a *Hilbert space*.

Proposition 5.12. *In a complex inner product space H, for all $h_1, h_2 \in H$:*

(i) ***Parallelogram Law:*** $\|h_1 + h_2\|^2 + \|h_1 - h_2\|^2 = 2(\|h_1\|^2 + \|h_2\|^2)$.

(ii) ***Polarization Identity:***

$$4(h_1, h_2) = \|h_1 + h_2\|^2 - \|h_1 - h_2\|^2 + i \left(\|h_1 + ih_2\|^2 - \|h_1 - ih_2\|^2 \right).$$

Crux. Expanding the norms using $\|x\|^2 = (x, x)$ yields:

$$\|h_1 + h_2\|^2 = \|h_1\|^2 + \|h_2\|^2 + (h_1, h_2) + (h_2, h_1)$$
$$\|h_1 - h_2\|^2 = \|h_1\|^2 + \|h_2\|^2 - (h_1, h_2) - (h_2, h_1)$$

Adding these equations proves (i). Subtracting them yields $\|h_1 + h_2\|^2 - \|h_1 - h_2\|^2 = 2(h_1, h_2)+2(h_2, h_1)$. Replacing h_2 with ih_2 and using conjugate linearity yields $\|h_1 + ih_2\|^2 - \|h_1 - ih_2\|^2 = -2i(h_1, h_2) + 2i(h_2, h_1)$. Multiplying this second result by i and adding it to the first isolates $4(h_1, h_2)$, proving (ii). □

Remark 5.13 (Fréchet-von Neumann-Jordan Theorem) A Banach space is a Hilbert space (i.e., its norm is induced by an inner product) if and only if its norm satisfies the Parallelogram Law.

Crux. Define a candidate inner product via the Polarization Identity. The Parallelogram Law guarantees additivity $(h_1 + h_2, h_3) = (h_1, h_3) + (h_2, h_3)$. Homogeneity $(ch_1, h_2) = c(h_1, h_2)$ is extended from $\mathbb{Q}$ to $\mathbb{C}$ via the continuity of the norm. Norm continuity is guaranteed by the Reverse Triangle Inequality: $|\|x\| - \|y\|| \leq \|x - y\|$. Thus, $x_n \to x \implies \|x_n\| \to \|x\|$.

□

Definition: Orthogonal Functions & Correlation Functions $f, g \in L^2(E, \mathbb{C})$ are *orthogonal* if $(f, g) = 0$. The Cauchy-Schwarz inequality guarantees $-1 \leq \rho \leq 1$, where:

$$\rho \triangleq \frac{(f, g)}{\|f\|_2 \|g\|_2}.$$

Geometrically, $\rho = \cos\theta$ defines the angle between f and g. If f and g are zero-mean random variables, ρ is their correlation.

Definition: Subspace A non-empty subset $K \subseteq H$ is a *subspace* if it is closed under vector addition and scalar multiplication. If K is also complete with respect to the induced norm, it is a *complete (or closed) subspace.*

Theorem 5.15. *(Orthogonal Projection) Let K be a complete subspace of a Hilbert space H. For each $h \in H$, there is a unique $h' \in K$ such that $h - h'$ is orthogonal to every element of K. Equivalently, h' is the unique minimizer of the distance to K: $\|h - h'\| = \inf_{k \in K} \|h - k\|$.*

Crux. Let $d \triangleq \inf_{k \in K} \|h - k\|$. We proceed in three steps:

(1) **Existence:** Choose a sequence $\{k_n\} \subset K$ such that $\|h - k_n\| \to d$. Applying the Parallelogram Law to $h - k_n$ and $h - k_m$:

$$\|2h - (k_n + k_m)\|^2 + \|k_n - k_m\|^2 = 2\|h - k_n\|^2 + 2\|h - k_m\|^2.$$

Since $(k_n + k_m)/2 \in K$, we know $\|h - (k_n + k_m)/2\|^2 \geq d^2$. Substituting this gives $4d^2 + \|k_n - k_m\|^2 \leq 2\|h - k_n\|^2 + 2\|h - k_m\|^2$. As $n, m \to \infty$, the RHS approaches $4d^2$, forcing $\|k_n - k_m\| \to 0$. The sequence is Cauchy in the complete space K, so it converges to some $h' \in K$. By continuity of the norm, $\|h - h'\| = d$.

(2) **Orthogonality:** For any $k' \in K$ and scalar t, $h' + tk' \in K$. A standard variational argument on the quadratic $f(t) = \|h - (h' + tk')\|^2 \geq d^2$ requires the first-order cross term to vanish, yielding the orthogonality condition $(h - h', k') = 0$.

(3) **Uniqueness:** If $(h - h', k') = 0$ for all $k' \in K$, then for any generic $k \in K$, the Pythagorean identity gives $\|h - k\|^2 = \|h - h'\|^2 + \|h' - k\|^2 \geq \|h - h'\|^2$, proving h' is the unique global minimizer.

□

Definition 5.17: The Space $L^p(E)$. For $1 \leq p < \infty$, the space $\mathcal{L}^p(E, \mathbb{C})$ consists of all measurable functions f such that $\int_E |f|^p \, \mathrm{d}m < \infty$. We define the quotient space $L^p(E, \mathbb{C}) \triangleq \mathcal{L}^p(E, \mathbb{C}) / \equiv$ under the usual "equality a.e." equivalence relation. The L^p-norm is defined as:

$$\|f\|_p \triangleq \left(\int_E |f|^p \, \mathrm{d}m \right)^{1/p}.$$

Definition 5.18: Essential Supremum & $L^\infty(E)$. For a measurable function $f : E \to \mathbb{C}$, the *essential supremum* of $|f|$ is defined as its L^∞-norm:

$$\|f\|_\infty \triangleq \inf\{c \geq 0 : |f| \leq c \text{ a.e.}\}.$$

If $\|f\|_\infty < \infty$, f is *essentially bounded*. The space of all such functions is $\mathcal{L}^\infty(E, \mathbb{C})$, with the quotient space denoted $L^\infty(E, \mathbb{C})$.

Note: Proposition 3.15 easily implies $L^\infty(E)$ is a vector space. The triangle inequality (Minkowski's inequality) for $1 \leq p < \infty$, which proves $L^p(E)$ is a vector space, will be established shortly.

Lemma 5.20 (Young's Inequality) For any non-negative x, y and $\alpha, \beta \in (0, 1)$ with $\alpha + \beta = 1$:

$$x^\alpha y^\beta \leq \alpha x + \beta y.$$

Crux. The function $f(t) = \alpha + \beta t - t^\beta$ for $t \geq 0$ has a unique minimum at $t = 1$ where $f(1) = 0$. For $x > 0$, substituting $t = y/x$ into $f(t) \geq 0$ directly yields the inequality. The $x = 0$ case is trivially true. □

Theorem 5.21. *(Hölder's Inequality). If $1 \leq p, q \leq \infty$ such that $\frac{1}{p} + \frac{1}{q} = 1$ (with $1/\infty \triangleq 0$), then for $f \in L^p(E)$ and $g \in L^q(E)$, we have $fg \in L^1(E)$ and:*

$$\|fg\|_1 \leq \|f\|_p \, \|g\|_q \, .$$

Crux. **Case $p = 1, q = \infty$:** Since $|g| \leq \|g\|_\infty$ a.e., we have $|fg| \leq |f| \, \|g\|_\infty$ a.e. Integrating over E immediately yields $\|fg\|_1 \leq \|f\|_1 \, \|g\|_\infty$. (The $p = \infty, q = 1$ case is symmetric).

Case $1 < p, q < \infty$**:** If $\|f\|_p = 0$ or $\|g\|_q = 0$, then $f = 0$ or $g = 0$ a.e., and the inequality trivially holds. Otherwise, normalize by setting $h = \frac{f}{\|f\|_p}$ and $k = \frac{g}{\|g\|_q}$, ensuring $\|h\|_p = \|k\|_q = 1$. Applying Lemma 5.20 with $x = |h|^p$, $y = |k|^q$, $\alpha = 1/p$, and $\beta = 1/q$:

$$\left|h\overline{k}\right| = |h|\,|k| \le \frac{|h|^p}{p} + \frac{|k|^q}{q}.$$

Integrating over E yields $\|hk\|_1 \le \frac{1}{p}\|h\|_p^p + \frac{1}{q}\|k\|_q^q = \frac{1}{p} + \frac{1}{q} = 1$. Substituting h and k back and rearranging provides the desired result. $\square$

Corollary 5.22 (Cauchy-Schwarz Inequality) If $f, g \in L^2(E)$, then $|(f,g)| \le \|f\|_2 \|g\|_2$.

Crux. This is an immediate consequence of Hölder's inequality by setting $p = q = 2$, noting that $|(f,g)| = \left|\int_E f\overline{g}\,\mathrm{d}m\right| \le \int_E |f\overline{g}|\,\mathrm{d}m = \|fg\|_1$. $\square$

Theorem 5.23. *(Minkowski's Inequality) For $1 \le p \le \infty$ and $f, g \in L^p(E, \mathbb{C})$, we have $\|f+g\|_p \le \|f\|_p + \|g\|_p$.*

Crux. The cases $p = 1, \infty$ are trivial. For $1 < p < \infty$, choose q such that $1/p + 1/q = 1$, which implies $q(p-1) = p$. We expand the integrand using the triangle inequality: $|f+g|^p = |f+g|\,|f+g|^{p-1} \le |f|\,|f+g|^{p-1} + |g|\,|f+g|^{p-1}$. Applying Hölder's Inequality to both terms on the right with exponents p and q:

$$\begin{aligned}\int_E |f+g|^p\,\mathrm{d}m &\le \|f\|_p \left\|(f+g)^{p-1}\right\|_q + \|g\|_p \left\|(f+g)^{p-1}\right\|_q \\ &= \left(\|f\|_p + \|g\|_p\right)\left(\int_E |f+g|^{q(p-1)}\,\mathrm{d}m\right)^{1/q} \\ &= \left(\|f\|_p + \|g\|_p\right)\|f+g\|_p^{p/q}.\end{aligned}$$

Dividing both sides by $\|f+g\|_p^{p/q}$ yields $\|f+g\|_p^{p-p/q} = \|f+g\|_p \le \|f\|_p + \|g\|_p$. $\square$

Theorem 5.24. *(Riesz-Fischer Theorem for L^p) The space $L^p(E, \mathbb{C})$ is complete for $1 \le p \le \infty$.*

Crux. **Case** $1 < p < \infty$**:** Let $\{f_n\}$ be Cauchy in L^p. Extract a fast subsequence where $\left\|f_{N_{k+1}} - f_{N_k}\right\|_p < 1/2^k$. Let $g_k = \sum_{l=1}^k \left|f_{N_{l+1}} - f_{N_l}\right|$. By Minkowski's inequality, $\|g_k\|_p \le \sum_{l=1}^k 1/2^l < 1$. By the Monotone Convergence Theorem, $g_k \nearrow g$ implies $\int_E g^p\,\mathrm{d}m \le 1$. Thus, $g \in L^p(E)$ and $g < \infty$ a.e. This bounded sequence of partial sums guarantees the telescoping series $f_{N_1} + \sum(f_{N_{k+1}} - f_{N_k})$ converges pointwise a.e. to some limit f. Since $|f| \le |f_{N_1}| + g$, Minkowski's inequality ensures $f \in L^p(E)$. Since $|f - f_{N_k}|^p \le g^p \in L^1(E)$, the Dominated Convergence Theorem gives $\|f - f_{N_k}\|_p \to 0$. A subsequence converging in the norm forces the entire Cauchy sequence to converge to $f \in L^p(E)$.

Case $p = \infty$: Let $\{f_n\}$ be Cauchy in L^∞. Since $|f_n - f_m| \leq \|f_n - f_m\|_\infty$ a.e., the sequence is uniformly Cauchy outside a null set. It converges uniformly pointwise to an essentially bounded limit f, trivially ensuring $\|f_n - f\|_\infty \to 0$. □

Theorem 5.25. *If $m(E) < \infty$, then $L^q(E, \mathbb{C}) \subseteq L^p(E, \mathbb{C})$ for $1 \leq p \leq q \leq \infty$.*

Crux. The case $q = \infty$ is trivial since bounded functions integrated over finite sets are finite. For $q < \infty$, we observe pointwise that $|f|^p \leq \max(1, |f|^q) \leq 1 + |f|^q$. Integrating over E yields $\int_E |f|^p \, \mathrm{d}m \leq m(E) + \int_E |f|^q \, \mathrm{d}m$. Since $m(E) < \infty$ and $f \in L^q(E)$, the RHS is finite, meaning $f \in L^p(E)$. □

Alternative Crux (via Hölder's Inequality): Write $\int_E |f|^p \, \mathrm{d}m = \int_E |f|^p \cdot 1 \, \mathrm{d}m$. Apply Hölder's with conjugate exponents $r = q/p$ and $s = q/(q-p)$. The integral is bounded by $\left(\int_E |f|^q \, \mathrm{d}m\right)^{p/q} m(E)^{1-p/q} < \infty$.

Definition 5.26 & 5.28: Moments & Variance For a random variable $X \in L^n(\Omega)$, its n-th *moment* is $\mathbb{E}(X^n) = \int_{\mathbb{R}} x^n \, \mathrm{d}P_X(x)$. Let $\mu \triangleq \mathbb{E}(X)$. The n-th *central moment* is $\mathbb{E}[(X-\mu)^n] = \int_{\mathbb{R}} (x-\mu)^n \, \mathrm{d}P_X(x)$. The *variance* is the second central moment: $\mathrm{Var}(X) \triangleq \mathbb{E}[(X-\mu)^2] = \mathbb{E}(X^2) - \mu^2$. If X has density $f_X(x)$, the integrals are taken against $f_X(x)\mathrm{d}x$.

Proposition 5.27. *If $\mathbb{E}(|X|^n) < \infty$, then $\mathbb{E}(|X|^k) < \infty$ for all $k \leq n$.*

Crux. Direct consequence of Theorem 5.25 ($L^n \subseteq L^k$), which applies perfectly because the probability space has finite measure ($P(\Omega) = 1 < \infty$). □

Proposition 5.29 & 5.30. *(Moment Conversions) Moments of order n uniquely determine central moments of order n (and vice versa), relying only on lower-order moments $k \leq n$.*

Crux. Follows immediately from the linearity of expectation and the binomial expansions of $(X-\mu)^n$ and $X^n = [(X-\mu)+\mu]^n$. □

Lemma 5.31.5 (Linearity): $\mathbb{E}(aX+b) = a\mathbb{E}(X) + b$ and $\mathrm{Var}(aX+b) = a^2\mathrm{Var}(X)$. (Direct consequence of integral linearity).

Theorems 5.32, 5.33 & Higher Gaussian Moments Let $X \sim \mathcal{N}(\mu, \sigma^2)$ with density $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}$. Then:

(1) $\mathbb{E}(X) = \mu$

(2) $\mathrm{Var}(X) = \sigma^2$

(3) $\mathbb{E}[(X-\mu)^{2k-1}] = 0$ (Odd central moments vanish due to symmetry).

(4) $\mathbb{E}[(X-\mu)^{2k}] = 1 \cdot 3 \cdot 5 \cdots (2k-1)\sigma^{2k}$ (Even moments via integration by parts).

Crux (Mean & Variance). Assume the standard normal $Z \sim \mathcal{N}(0,1)$. The mean integral is an odd function, thus 0. The variance integral evaluates to 1 (Exercise 4.11). Applying the affine transformation $X = \sigma Z + \mu$ and Lemma 5.31.5 yields $\mathbb{E}(X) = \mu$ and $\mathrm{Var}(X) = \sigma^2$. □

Remark 5.35 (Characteristic Function & Moments) If the characteristic function $\varphi_X(t)$ is k-times continuously differentiable at $t = 0$, then X has a finite k-th moment, given by:

$$\mathbb{E}(X^k) = \frac{1}{i^k} \frac{\mathrm{d}^k}{\mathrm{d}t^k} \varphi_X(t) \bigg|_{t=0}.$$

Theorem 5.36. *The random variables X, Y are independent if and only if*

$$\mathbb{E}(f(X)g(Y)) = \mathbb{E}(f(X))\mathbb{E}(g(Y))$$

holds for all bounded Borel measurable functions f, g.

Crux. ($\Longleftarrow$) Let $f = \mathbb{I}_A$ and $g = \mathbb{I}_B$ for Borel sets $A, B \in \mathcal{B}$. The relation yields $P(X \in A, Y \in B) = P(X \in A)P(Y \in B)$, which is the exact definition of independence.

($\Longrightarrow$) Assume X, Y are independent. We build up the class of functions using standard measure-theoretic induction (the "Standard Machine"):

(1) **Indicators:** By definition of independence, the relation holds for $f = \mathbb{I}_A$, $g = \mathbb{I}_B$.

(2) **Simple Functions:** The relation extends to simple functions via the linearity of expectation.

(3) **Non-Negative Functions:** For any non-negative measurable functions $f, g \geq 0$, there exist sequences of simple functions $f_n \nearrow f$ and $g_n \nearrow g$. By the Monotone Convergence Theorem (which requires no boundedness), $\mathbb{E}(f_n(X)g_n(Y)) \to \mathbb{E}(f(X)g(Y))$.

(4) **General Functions:** Decompose arbitrary functions into their positive and negative parts: $f = f^+ - f^-$ and $g = g^+ - g^-$. Linearity extends the relation to all functions, provided the expectations exist.

Note: We do not actually need f and g to be bounded. The factorization holds for *any* measurable functions as long as the expectations $\mathbb{E}(|f(X)|)$, $\mathbb{E}(|g(Y)|)$, and $\mathbb{E}(|f(X)g(Y)|)$ exist. Boundedness is merely assumed in the theorem statement as a shortcut to guarantee these integrability conditions trivially. □

Proposition 5.37. *Let X, Y be independent random variables. If $\mathbb{E}(X) = 0$ and $\mathbb{E}(Y) = 0$, then $\mathbb{E}(XY) = 0$.*

Crux. First, we establish that $XY \in L^1$. Let $u_n(x) = |x| \mathbb{I}_{[-n,n]}(x)$ and $v_n(y) = |y| \mathbb{I}_{[-n,n]}(y)$. As these are bounded Borel functions, Theorem 5.36 gives $\mathbb{E}(u_n(X)v_n(Y)) = \mathbb{E}(u_n(X))\mathbb{E}(v_n(Y))$. Since $u_n(X)v_n(Y) \nearrow |XY|$ monotonically, the Monotone Convergence Theorem yields $\mathbb{E}(|XY|) = \mathbb{E}(|X|)\mathbb{E}(|Y|)$. Since $X, Y \in L^1$, their absolute expectations are finite, so $\mathbb{E}(|XY|) < \infty$.

Now, let $f_n(x) = x\mathbb{I}_{[-n,n]}(x)$ and $g_n(y) = y\mathbb{I}_{[-n,n]}(y)$. By Theorem 5.36, $\mathbb{E}(f_n(X)g_n(Y)) = \mathbb{E}(f_n(X))\mathbb{E}(g_n(Y))$. As $n \to \infty$, $f_n(X)g_n(Y) \to XY$ pointwise. Since the sequence is dominated by our integrable bound ($|f_n(X)g_n(Y)| \leq |XY| \in L^1$), the Dominated Convergence Theorem applies:

$$\mathbb{E}(XY) = \lim_{n\to\infty} \mathbb{E}(f_n(X)g_n(Y)) = \lim_{n\to\infty} \mathbb{E}(f_n(X)) \cdot \lim_{n\to\infty} \mathbb{E}(g_n(Y)) = \mathbb{E}(X)\mathbb{E}(Y) = 0.$$

□

Definition 5.38: Covariance & Correlation For $X \in L^1(\Omega)$ with mean $\mu_X = \mathbb{E}(X)$, the *centered* random variable is $X_c \triangleq X - \mu_X$, so $\mathbb{E}(X_c) = 0$. For $X, Y \in L^2(\Omega)$, the *covariance* is:

$$\mathrm{Cov}(X, Y) \triangleq \mathbb{E}(X_c Y_c) = \mathbb{E}(XY) - \mu_X \mu_Y.$$

The *correlation* is the inner product of the normalized, centered variables:

$$\rho_{X,Y} \triangleq \frac{\mathrm{Cov}(X, Y)}{\|X_c\|_2 \|Y_c\|_2} = \frac{\mathrm{Cov}(X, Y)}{\sqrt{\mathrm{Var}(X)\mathrm{Var}(Y)}}.$$

X and Y are *uncorrelated* if $\rho_{X,Y} = 0$.

ERRATA (Capinski & Kopp): The textbook incorrectly defines the denominator of $\rho_{X,Y}$ as $\|X\|_2 \|Y\|_2$ (the uncentered L^2-norms). This is mathematically false unless $\mu_X = \mu_Y = 0$. The correct denominator strictly requires the centered norms $\|X_c\|_2 \|Y_c\|_2$, which represent the standard deviations $\sigma_X \sigma_Y$. Without centering, the Cauchy-Schwarz inequality fails to bound the correlation strictly within the interval $[-1, 1]$.

Proposition 5.40. *If $\{X_i\}_{i=1}^n$ are pairwise uncorrelated, then* $\mathrm{Var}\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n \mathrm{Var}(X_i)$.

Crux. By induction, it suffices to prove this for $n = 2$. Using the linearity of expectation on the centered variables $(X + Y)_c = X_c + Y_c$:

$$\mathrm{Var}(X + Y) = \mathbb{E}([(X + Y)_c]^2) = \mathbb{E}((X_c + Y_c)^2) = \mathbb{E}(X_c^2) + \mathbb{E}(Y_c^2) + 2\mathbb{E}(X_c Y_c).$$

Since X and Y are uncorrelated, $\mathbb{E}(X_c Y_c) = \mathrm{Cov}(X, Y) = 0$. The remaining terms are exactly $\mathrm{Var}(X) + \mathrm{Var}(Y)$. □

Proposition 5.41. *If $\{X_i\}_{i=1}^n$ are independent, the characteristic function of their sum is the product of their characteristic functions:*

$$\varphi_{\sum X_i}(t) = \prod_{i=1}^n \varphi_{X_i}(t).$$

Crux. By induction, it suffices to prove this for $n = 2$. Let $f(x) = e^{itx}$ and $g(y) = e^{ity}$. Since $|f(x)| = |g(y)| = 1$, they are bounded measurable functions. Extending Theorem 5.36 to complex-valued functions (via linearity over real and imaginary parts), independence implies:

$$\varphi_{X+Y}(t) = \mathbb{E}(e^{it(X+Y)}) = \mathbb{E}(f(X)g(Y)) = \mathbb{E}(f(X))\mathbb{E}(g(Y)) = \varphi_X(t)\varphi_Y(t).$$

□

Conditional Expectation: Geometric Motivation Theorem 5.15 establishes that we can orthogonally project any vector in a Hilbert space onto a complete subspace. If $\mathcal{G}$ is a sub-σ-field of $\mathcal{F}$, then the space of $\mathcal{G}$-measurable functions, $L^2(\mathcal{G})$, is a closed (and thus complete) subspace of $L^2(\mathcal{F})$. For any random variable $X \in L^2(\mathcal{F})$, its orthogonal projection onto $L^2(\mathcal{G})$ is a unique random variable $Y \in L^2(\mathcal{G})$. By definition of orthogonality, the error $X - Y$ is orthogonal to every $Z \in L^2(\mathcal{G})$. Taking $Z = \mathbb{I}_G$ for any $G \in \mathcal{G}$, the inner product $(X - Y, \mathbb{I}_G) = 0$ yields:

$$\int_G Y \, \mathrm{d}P = \int_G X \, \mathrm{d}P. \tag{5.5}$$

Definition 5.42: Conditional Expectation Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{G}$ a sub-σ-field of $\mathcal{F}$ (refer to Definition 2.30(ii)). For an $\mathcal{F}$-measurable X, if there exists a unique (a.e.) $\mathcal{G}$-measurable Y such that 5.5 holds for all $G \in \mathcal{G}$, we call Y the *conditional expectation* of X given $\mathcal{G}$, denoted $Y = \mathbb{E}(X|\mathcal{G})$.

Note: While the geometric motivation strictly requires L^2 spaces to utilize the Orthogonal Projection Theorem, the integral condition in 5.5 only requires the functions to be in L^1. The following theorem guarantees that $\mathbb{E}(X|\mathcal{G})$ exists for all integrable random variables, not just those with finite variance.

Theorem 5.43. *(First Construction of Conditional Expectation) Let $(\Omega, \mathcal{F}, P)$ be a probability space and $\mathcal{G} \subseteq \mathcal{F}$ a sub-σ-field. For every $X \in L^1(\mathcal{F})$, the conditional expectation $Y = \mathbb{E}(X|\mathcal{G})$ exists and is unique in $L^1(\mathcal{G})$.*

Crux. We construct Y iteratively using standard measure-theoretic induction:

(1) **Bounded X:** If X is essentially bounded ($X \in L^\infty$), then $X \in L^2$ (since $P(\Omega) = 1 < \infty$). By the Orthogonal Projection Theorem (5.15), a unique projection $Y \in L^2(\mathcal{G}) \subseteq L^1(\mathcal{G})$ exists satisfying 5.5.

(2) **Non-negative $X \geq 0$:** Let $X_n = \min(X, n)$. Since X_n is bounded, $Y_n = \mathbb{E}(X_n|\mathcal{G})$ exists. Because $X_n \nearrow X$, the integral condition ensures $\int_G Y_n \, \mathrm{d}P \leq \int_G Y_{n+1} \, \mathrm{d}P$ for all $G \in \mathcal{G}$. Testing this inequality on the specific set $G = \{Y_n > Y_{n+1}\} \in \mathcal{G}$ forces $P(G) = 0$, meaning Y_n is monotonically increasing a.e. Let $Y = \lim_{n\to\infty} Y_n$. Applying the Monotone Convergence Theorem to both sides of $\int_G Y_n \, \mathrm{d}P = \int_G X_n \, \mathrm{d}P$ yields $\int_G Y \, \mathrm{d}P = \int_G X \, \mathrm{d}P < \infty$. Thus $Y \in L^1(\mathcal{G})$.

(3) **Arbitrary $X \in L^1$:** Decompose $X = X^+ - X^-$. The conditional expectations for X^+ and X^- exist by step (2). Linearity of the integral gives $Y = \mathbb{E}(X^+|\mathcal{G}) - \mathbb{E}(X^-|\mathcal{G})$.

(4) **Uniqueness:** If $Z \in L^1(\mathcal{G})$ also satisfies 5.5, then $\int_G (Y - Z)\,\mathrm{d}P = 0$ for all $G \in \mathcal{G}$. Taking $G = \{Y > Z\}$ and $G = \{Y < Z\}$ forces $Y = Z$ a.e.

$\square$